# CLASS 6 — MATHS · FORMULA SHEET

Chapter 7: **Fractions** | Quick revision, one page

## Kinds of fractions

- Fraction $\frac{a}{b}$: numerator $a$ (parts taken), denominator $b$ (equal parts in the whole).
- Proper: $a < b$ (less than 1). Improper: $a \geq b$ (1 or more). Mixed number: whole part plus fraction.
- Mixed to improper: $c\frac{a}{b} = \frac{c \times b + a}{b}$. Improper to mixed: divide the numerator by the denominator.

## Equivalent fractions

- $\frac{a}{b} = \frac{a \times k}{b \times k} = \frac{a \div k}{b \div k}$ for any non-zero $k$.
- Simplest form: divide numerator and denominator by their HCF.

## Comparing

- Same denominator: the larger numerator is greater. Same numerator: the smaller denominator is greater.
- Unlike fractions: change to a common denominator (LCM of denominators), then compare.

## Addition and subtraction

- Like fractions: $\frac{a}{c} + \frac{b}{c} = \frac{a+b}{c}$ and $\frac{a}{c} - \frac{b}{c} = \frac{a-b}{c}$ (denominator unchanged).
- Unlike fractions: make denominators equal first. Mixed numbers: add or subtract the whole parts and the fractions, or convert to improper fractions.

## Fraction of a quantity

- $\frac{a}{b}$ of $N = \frac{N}{b} \times a$ (divide into $b$ equal parts, take $a$).
- Remaining part $= 1 -$ fraction used.

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